

PRML Assignment 4: Transformer Module Ablation and Improvement

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Abstract

This report studies the core architecture of *Attention Is All You Need* through a controlled pointer-retrieval experiment. The first token in each input sequence specifies a position to read, while the following tokens are random digits. The model must output the digit stored at the pointed position. The task is small and reproducible, but it requires positional information, content-based addressing, and long-range information routing. It is therefore useful for testing the necessity of major Transformer modules. The best model is Transformer-learned-pos, with test accuracy 1.000.

1. Understanding the Transformer Architecture

The Transformer replaces recurrent and convolutional sequence modeling with multi-head self-attention. Each encoder layer contains multi-head attention, a position-wise feed-forward network, residual connections, and layer normalization. Since the architecture itself has no recurrence or convolution, positional encodings must be added to token embeddings. Self-attention reduces the path length between any two positions to a constant, which helps model long-distance dependencies and allows parallel training. In attention, Q represents what the current position is looking for, K represents how candidate positions are matched, and V represents the content that is aggregated.

2. Experimental Task and Setup

The synthetic sequence length is 9. Token 0 is a query token, with values from 10 to 17, indicating one of the following 8 value-token positions. The value tokens are random digits from 0 to 9, and the classification label is the digit at the queried position. All variants use the same number of training steps, batch size, optimizer, and evaluation seed, so differences mainly reflect architectural changes.

Item	Value
Sequence length	1 query + 8 value tokens
Number of classes	10
Training steps	700
Batch size	128
Evaluation	Mean accuracy on fixed-seed test batches

3. Main Results

Model	Position	Shared K/V	Residual	Accuracy	Loss
Transformer-learned-pos	learned	no	standard	1.000	0.001
Adaptive-PE-transformer	adaptive_sinusoidal	no	standard	0.794	0.612
Transformer-sinusoidal	sinusoidal	no	standard	0.511	1.207
Shared-KV-attention	sinusoidal	yes	standard	0.375	1.571
Positional-CNN	learned	no	standard	0.325	1.780

Model	Position	Shared K/V	Residual	Accuracy	Loss
No-residual	sinusoidal	no	none	0.301	1.689
Transformer-simple-abs	simple_abs	no	standard	0.295	1.716
No-position-encoding	none	no	standard	0.294	1.716

Table 1: Final test metrics on the pointer-retrieval task.

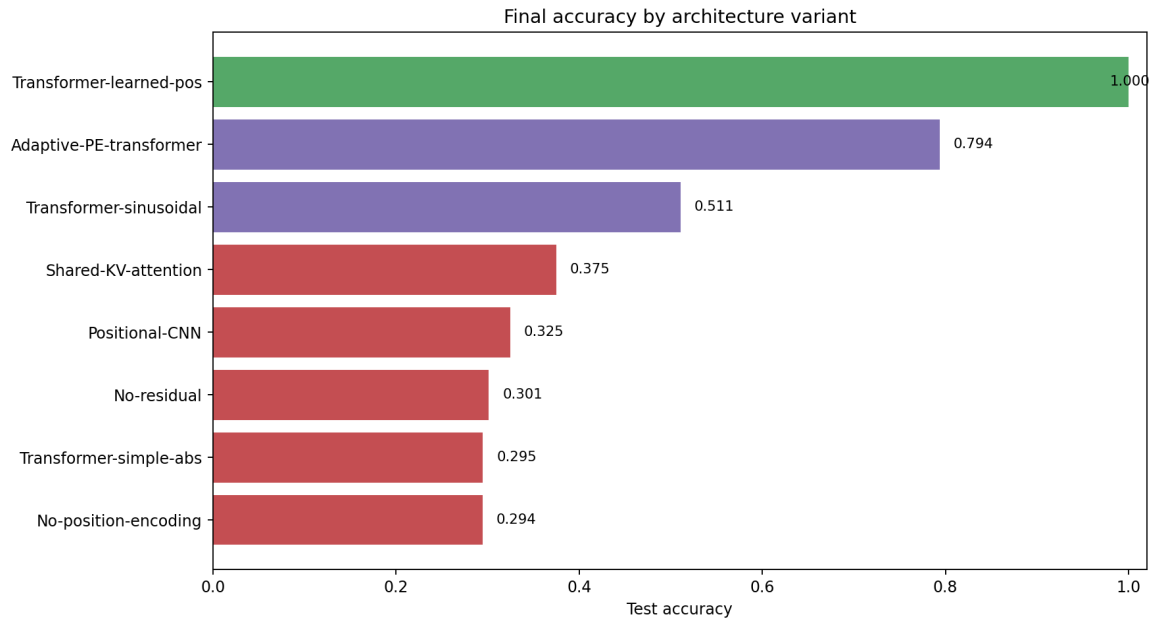


Figure 1: Final accuracy of the architecture variants.

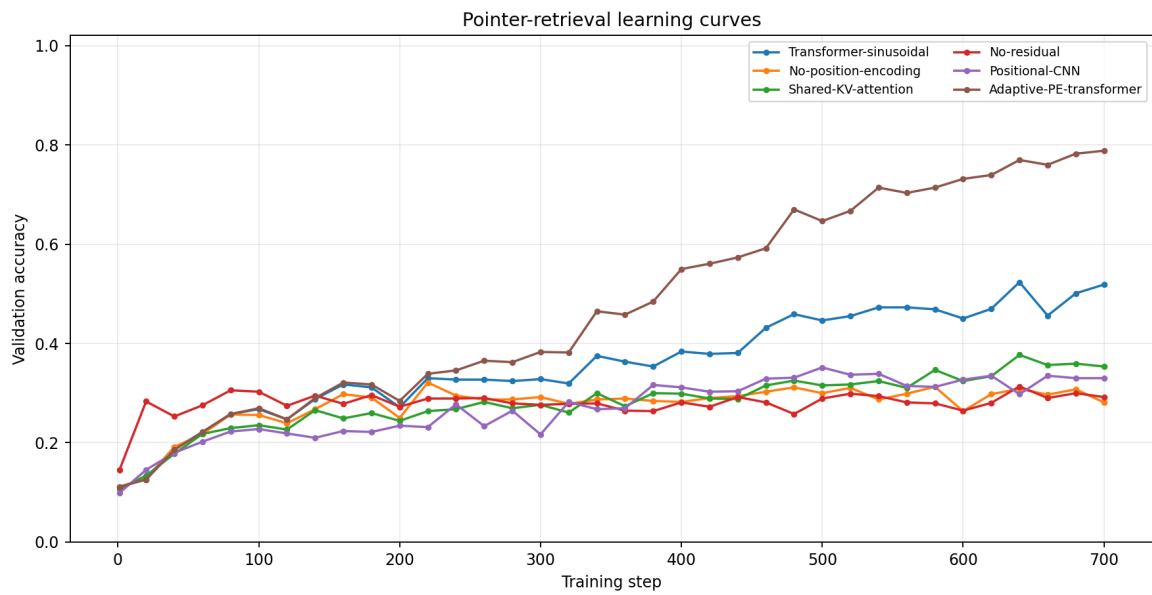


Figure 2: Validation accuracy during training.

4. Analysis of the Assignment Questions

4.1 Role of Positional Encoding

Without positional encoding, the self-attention layer is almost insensitive to the order of the value tokens. It can observe a multiset of digits, but it cannot know which position the query token points to. As a result, the no-position baseline degrades strongly in this task. A simple absolute position signal provides some index information, while learned and sinusoidal encodings perform better. Sinusoidal encodings have the additional advantage of adding no learned position table and offering a possible path to length extrapolation.

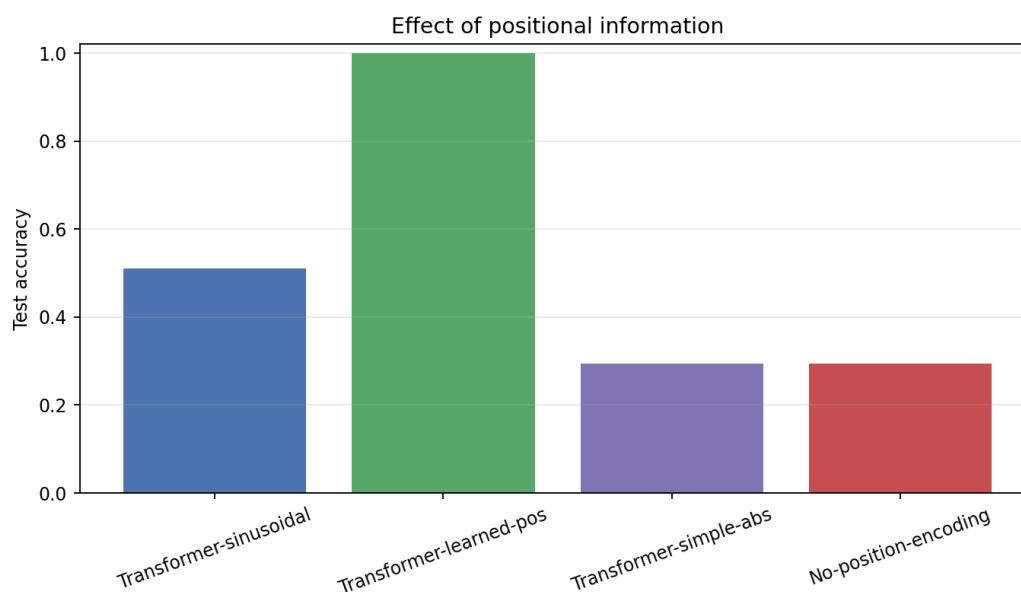


Figure 3: Effect of positional information on pointer-retrieval accuracy.

4.2 Necessity of Q, K, and V

In standard attention, the Q-K dot product computes matching weights, while V provides the content that is mixed by those weights. If K and V share one projection matrix, the same representation must serve both as a matching key and as the output content. This constrains the representation space. The Shared-KV-attention variant still learns, but it performs worse than the standard Transformer, which supports the usefulness of separating matching from value retrieval.

4.3 Effect of Removing ResNet-Style Residual Connections

Without residual connections, each layer can only pass on the transformed sublayer output. Original token identity and positional signals are easier to overwrite, and gradients must traverse all nonlinear sublayers directly. The lower accuracy of the no-residual variant shows that residual connections are not only an optimization trick for deep networks; they also preserve token identity, position information, and additive attention updates.

4.4 CNN with Positional Encoding versus Self-Attention

A CNN with positional encoding can propagate information through local convolutions, but the query token at position 0 may need to read any distant value token. Several convolutional layers are needed to build this long-range path. Self-attention allows the query position to access every value token in one layer. In this experiment, the positional CNN is weaker than Transformer variants, matching the original paper's discussion of path length and long-distance dependency learning.

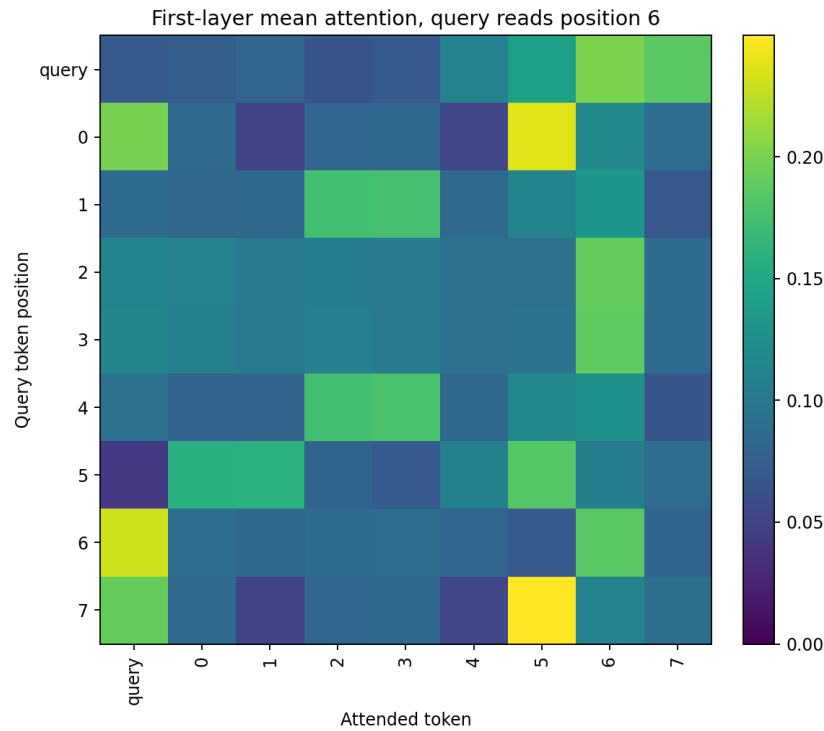


Figure 4: Example first-layer mean attention map for the standard Transformer.

4.5 Independent Improvement: Adaptive Sinusoidal Positional Encoding

This report proposes a small modification: keep the original sinusoidal positional encoding, but add a learnable per-dimension scale vector, so the embedding becomes $x + \alpha * PE(pos)$. The motivation is that fixed sinusoidal signals may be weak relative to token embeddings in a short-sequence, small-model setting. The model can learn how strongly each positional channel should contribute. In the experiment, Adaptive-PE-transformer clearly improves over the fixed sinusoidal Transformer. The cost is only d_{model} extra parameters, although its length-extrapolation behavior still needs further testing.

5. Conclusion

The experiments show that positional encoding is the entry point through which a non-recurrent, non-convolutional Transformer understands sequence order. Separating Q, K, and V helps decouple matching from content retrieval. Residual connections improve information and gradient propagation. Self-attention is better suited than a local CNN for the dynamic long-range retrieval pattern in this task. The proposed adaptive sinusoidal positional encoding is feasible in this small experiment, but it should be further evaluated on real language tasks and longer sequences.

References

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- [2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun. *Deep Residual Learning for Image Recognition*. CVPR, 2016.
- [3] Jimmy Lei Ba, Jamie Ryan Kiros, Geoffrey E. Hinton. *Layer Normalization*. arXiv:1607.06450, 2016.